PH4102: Introductory Astrophysics

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Stellar evolution

Once a star has settled onto the main sequence, its luminosity changes very little until it begins to move off the main sequence when the helium core has mass about 10% of the mass of the star. What then happens to the star after this? Unfortunately, the only way of answering this question is through very detailed numerical computations. Nothing much can be done analytically or on the basis of general arguments.

There are important differences between the way in which *low* and *high* mass stars reach this point in their evolution which affects their subsequent evolution.

The low mass star (approximately within 0.3 $M_{\odot}$ to 1.2 $M_{\odot}$) possess radiative cores, whereas high mass stars possess convective cores.

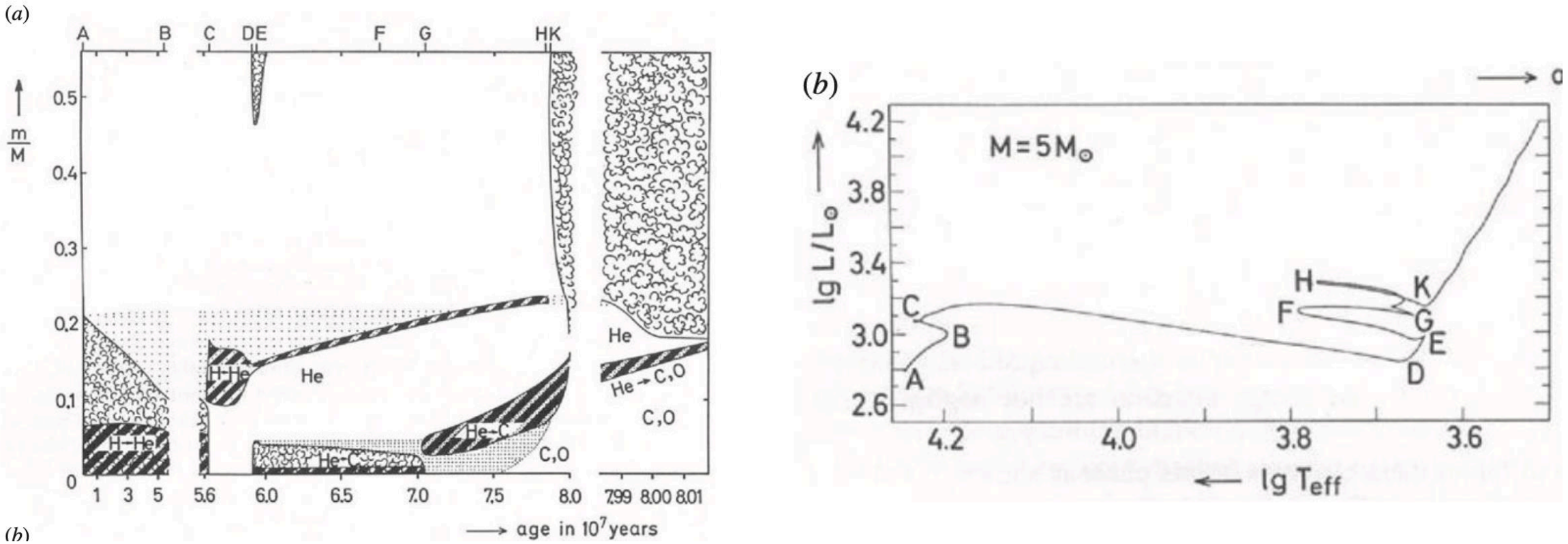
Therefore, in *high mass stars*, the transport of energy by convection in the central regions results in unprocessed material being continually convected into the nuclear burning regions and so the hydrogen abundance decreases uniformly within these regions until the hydrogen is exhausted.

In contrast, in *low mass stars*, the size of the hydrogen-burning zone increases gradually outwards with time until 10% of the mass is in a central helium core.

Thus, the exhaustion of the fuel in the core is rather more gentle in the case of low mass stars as compared with those in convective equilibrium.

Evolution track of a high mass star:

Evolution of a 5 $M_{\odot}$ star. Remember, the CNO cycle dominates for such stars.



From *High Energy Astrophysics*, M S Longair

The heavily hatched areas indicate the regions in which there is large nuclear energy production. The evolution proceeds as follows:

1. At the point A, the star begins its lifetime on the main sequence. The convective core contains 21% of the mass of the star and nuclear burning takes place within the inner 7% by mass. During the first 5.6×10^7 years, the star remains at roughly the same location on the H-R diagram, evolving to the point B.

2. By the point C, the central hydrogen fuel is exhausted and during the transition from C to D, an isothermal helium core is formed which begins to collapse, accompanied by the rapid expansion of the envelope to form a giant star. During the evolution from C to D, hydrogen burning continues in a shell about the helium core. At the point D, an outer convection zone is formed in the giant envelope.

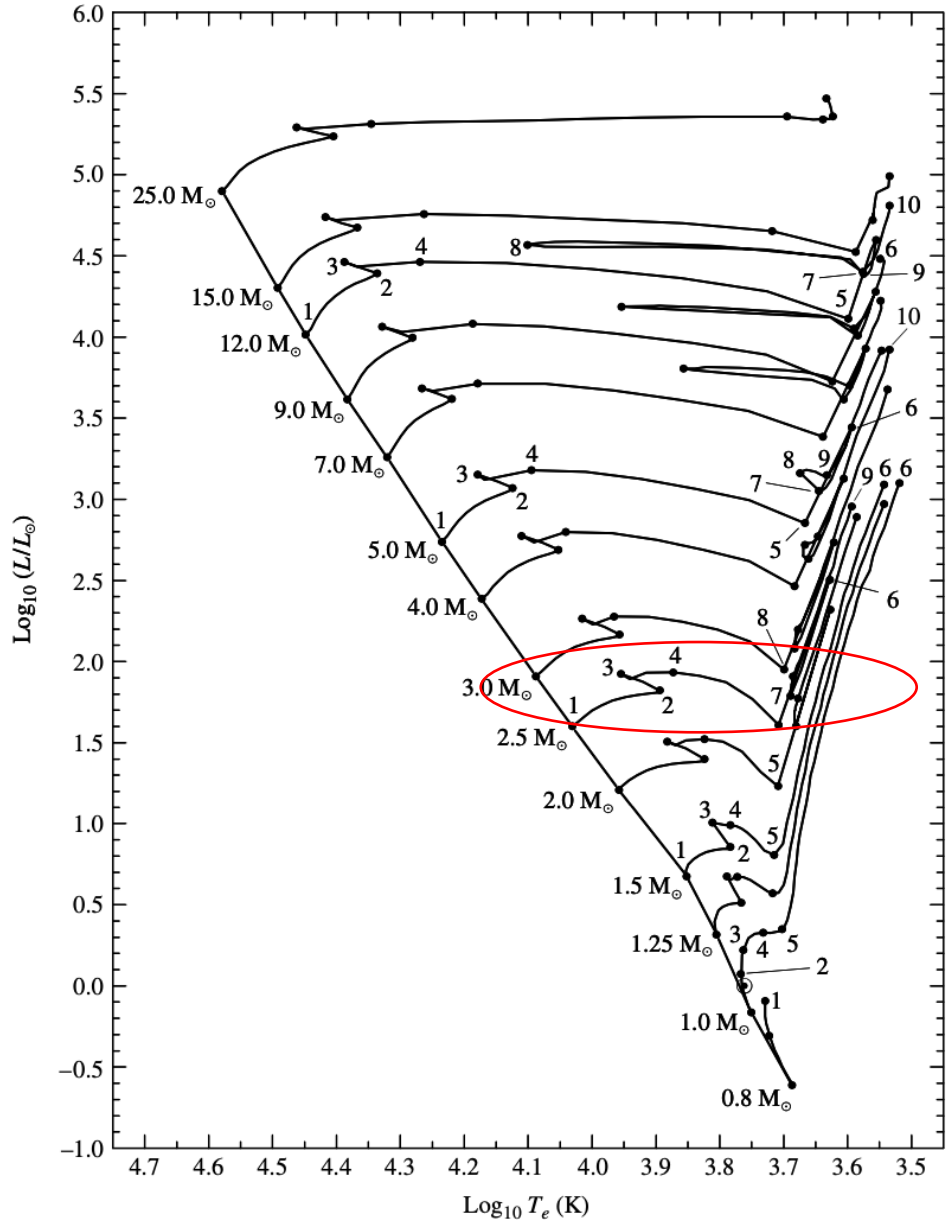
3. The continuing contraction of the central regions heats up the core until helium burning takes place at E. In the helium burning process, helium is converted into carbon $3\ ^4He \rightarrow ^{12}C$ through the rare triple-α process. This is accompanied by an excursion to higher temperatures across the H-R diagram to F.

4. Helium burning continues until the central helium abundance is reduced to zero and an isothermal ^{12}C core forms at G. Helium burning continues in a shell about the isothermal C,O core.

5. Throughout the stages D to H, hydrogen shell burning continues to larger and larger radii, but at H hydrogen shell burning ends because the temperature in the envelope is too low.

6. At K, the star develops a deep outer convection zone and subsequently moves almost vertically up in the H-R diagram.

In yet more massive stars, post-main sequence evolution proceeds by successive core and shell burning to produce nuclei with higher and higher binding energies (till iron.)



Numbers in the table are also shown in Figure. They represent elapsed times since main sequence in unit of millions of years (Myr).

Initial Mass ($M_\odot$)	1 6	2 7	3 8	4 9	5 10
25	0 6.51783	6.33044 7.04971	6.40774 7.0591	6.41337	6.43767
15	0 11.6135	11.4099 11.6991	11.5842 12.7554	11.5986	11.6118
12	0 16.1150	15.7149 16.4230	16.0176 16.7120	16.0337 17.5847	16.0555 17.6749
9	0 26.5019	25.9376 27.6446	26.3886 28.1330	26.4198 28.9618	26.4580 29.2294
7	0 43.4304	42.4607 45.3175	43.1880 46.1810	43.2291 47.9727	43.3388 48.3916
5	0 95.2108	92.9357 99.3835	94.4591 100.888	94.5735 107.208	94.9218 108.454
4	0 166.362	162.043 172.38	164.734 185.435	164.916 192.198	165.701 194.284
3	0 357.310	346.240 366.880	352.503 420.502	352.792 440.536	355.018
2.5	0 595.476	574.337 607.356	584.916 710.235	586.165 757.056	589.786
2	0 1148.10	1094.08 1160.96	1115.94 1379.94	1117.74 1411.25	1129.12
1.5	0 2910.76	2632.52	2690.39	2699.52	2756.73
1.25	0 5588.92	4703.20	4910.11	4933.83	5114.83
1	0 12269.8	7048.40	9844.57	11386.0	11635.8
0.8	0	18828.9	25027.9		

Main-sequence and post-main-sequence evolutionary tracks of stars with an initial composition of X = 0.68, Y = 0.30, and Z = 0.02 on a theorists' H-R diagram.

For stars less massive than the Sun, the central temperatures are lower and their luminosities correspondingly smaller. Eventually, at a low enough mass, the central temperature is not sufficiently high for the nuclear reactions of the p-p chain to take place, the corresponding mass being about 0.08 $M_{\odot}$.

Objects with masses in the range 0.08 $M_{\odot} > M > 0.01\ M_{\odot}$ are referred to as *brown dwarfs* and these are very faint infrared objects.

Objects with masses in the range $M < 0.01\ M_{\odot}$ are usually referred to as *exoplanets* (the mass of Jupiter being 0.001 $M_{\odot}$).

Mass loss:

An important part of the evolution is the phenomenon of *mass loss* from the outer envelopes of stars. Stars lose mass from their surfaces throughout much of their lives. Solar Wind is an example of *quiescent mass loss*. The solar wind is an example of thermally driven wind. It is caused by the high temperature of the corona, which makes it difficult for gravity to hold on to the gas.

The radiation force in the outer atmosphere of a very massive star may become comparable to gravity. This may cause a radiatively driven wind.

If a star is rotating very fast, that may lead to a centrifugally driven wind.

There are, however, much more violent forms of mass loss which are associated with the various evolutionary changes which stars undergo both when they are on the main sequence and after they have left it.

Because of the weak gravity at the surface of a red giant, often red giants have much stronger winds. It is possible for a star to lose a significant fraction of its mass while passing through the red giant phase.

Confirmation of mass loss comes from the observations of what are called *planetary nebulae*. Through the low-resolution telescopes of earlier times, a planetary nebula looked somewhat like a planet. We now know that a planetary nebula is essentially the *outer shell of a star which has been blown off*. At the centre of a planetary nebula, we usually find the hot core of the star, which is eventually expected to become a white dwarf.



Ring nebula by JWST

There is an even more violent mass loss mechanism known as *supernova*.

Supernova is so bright that it becomes visible during daytime. In fact because of this, we now know of their observations since thousand years (since 1054!).

No supernova has been observed in our Galaxy after the invention of the telescope! However, a supernova was seen in 1987 (named 1987A) in the Large Magellanic Cloud, which is a companion to our Galaxy at a distance of about 55 kpc.

The energy involved in a typical supernova explosion is estimated to be about 10^{45} J.

Supernovae can be divided into two types: Type I supernovae and Type II supernovae (with further sub-division).

Out of these, Type Ia, are believed to be produced as a result of catastrophic explosion of white dwarf stars. Such a situation may arise in a binary with a white dwarf and a red giant. Then, it is possible for a mass transfer to take place onto the white dwarf through gravitation capture of mass from the outer part of red giant. Now because of this mass acquisition, if the white dwarf mass exceeds it's limit (1.4 $M_{\odot}$ known as Chandrasekhar mass limit), the explosion may happen.

Type II supernovae are believed to take place in much more massive stars. When the core of the massive star completely runs out of all nuclear fuels, it starts shrinking until the core density becomes comparable to the density inside an atomic nucleus ($\approx 10^{17}$ kg m−3). When this happens, the rapidly shrinking core suddenly stops shrinking any more because neutron degeneracy pressure may balance gravity at such densities. The surrounding material falling inward with the core gets bounced back when the collapse of the core is suddenly halted. Presumably the Type II supernova is caused by the explosive bouncing off of the envelope surrounding the newly formed neutron star core. Our current view is that the elements heavier than iron are synthesized in Type II supernovae, in a way suggested in a classic paper by Burbidge, Burbidge, Fowler and Hoyle (1957). 1987A is an example of type II supernova.